Sentiment Classification of Customer-Service Conversations Using Fine-Tuned BERT with Tabular Feature Fusion

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Abstract*— *This study investigates three-class sentiment classification (positive, negative, neutral) of customer-service conversations using a fine-tuned BERT model augmented with learned embeddings for structured metadata. The training set comprises 970 samples and is highly imbalanced, with only 17 positive examples. To mitigate this imbalance, Gemini-based paraphrasing and English-French-English back-translation are utilized, increasing the number of positive samples to 68. Weighted cross-entropy loss and differential learning rates are implemented during fine-tuning. The proposed BertTabular model achieves a best validation macro-F1 score of 0.862 and a test macro-F1 score of 0.614. All experiments are tracked using Weights & Biases under project DI725-sentiment, run bert-tabular-v1.

I. Introduction

Sentiment analysis of customer-support interactions allows businesses to monitor service quality at scale. In contrast to product reviews, support conversations are multi-turn dialogues that include technical jargon, polite filler, and evolving sentiment, which complicates classification. Due to the limited size of the training set (970 labelled samples), training a transformer from scratch is impractical. Therefore, bert-base-uncased is fine-tuned, and its representations are enriched with lightweight embeddings for structured metadata fields. The resulting model is referred to as BertTabular.

II. Dataset and Pre-Processing

A. Dataset

The dataset contains customer-support dialogues labelled as neutral, negative, or positive. The training split has 970 samples and the test split has 30, with no missing values in either. Each sample includes 11 features: the raw conversation text and structured fields such as issue area, issue category, product category, agent experience level, and issue complexity.

B. Exploratory Data Analysis

Cramer's V quantified the association between each categorical feature and the sentiment target. The issue_sub_category feature scored V = 1.00, indicating data leakage, and was therefore removed. Issue_category (V = 0.739) and product_sub_category (V = 0.213) were retained as informative features. The target distribution is highly imbalanced: neutral = 542, negative = 411, positive = 17 (1.75% of training data)

Class	***Original***	***Augmented***
Neutral	542	542
Negative	411	411
Positive	17	68
Total	970	1021

Table I. Class distribution before and after augmentation.

C. Feature Selection

The following columns were dropped:

- issue_category_sub_category (redundant concatenation)
- agent_experience_level_desc (verbose description)
- issue_sub_category (V = 1.00, data leakage)
- agent_experience_level
- product_category
- issue_complexity (V < 0.08)

Retained features:

- conversation
- issue_area
- issue_category
- product_sub_category

D. Tokenisation

BERT supports a maximum input length of 512 tokens. Conversations in the dataset range from 46 to 5,708 characters, with some exceeding this token limit. Because the closing portion of a conversation typically contains more sentiment information than the opening, long sequences are truncated from the beginning, preserving the final 510 tokens in addition to the [CLS] and [SEP] tokens.

E. Class Imbalance

Two augmentation strategies were applied to the 17 positive samples. First, Gemini 1.5 Flash paraphrased each conversation while preserving speaker roles and sentiment, resulting in 34 samples. Second, English-to-French-to-English back-translation using MarianMT was applied to these 34 samples, yielding a total of 68 positive samples. Balanced class weights (neutral: 0.628, negative: 0.828, positive: 5.005) were incorporated into the cross-entropy loss function.

III. Modeling

BertTabular integrates two representation streams. The first stream is the [CLS] pooler output from bert-base-uncased (768 dimensions). The second stream consists of learned embeddings (8 dimensions each) for the three retained categorical features, concatenated into a 24-dimensional vector. The combined 792-dimensional representation is processed through a sequence of layers: a linear transformation from 792 to 256 dimensions, ReLU activation, dropout (0.3), and a final linear transformation from 256 to 3 dimensions.

The model was trained for six epochs with a batch size of 16 using the AdamW optimizer (weight decay 0.01). Differential learning rates were applied: 2e-5 for the BERT backbone to prevent catastrophic forgetting, and 1e-3 for the classification head and categorical embeddings. A linear warmup over 10% of steps was followed by linear decay. The dataset was split into 867 training and 154 validation samples using stratified sampling.

IV. Evaluation Metrics

Macro-averaged F1 is used as the primary evaluation metric. It computes F1 for each class independently and averages the results, assigning equal weight to each class regardless of support, which is essential given the class imbalance. Accuracy is reported as a secondary metric but is not used for model selection, since predicting only the neutral class would yield 55.7% accuracy. Per-class precision and recall are also reported to diagnose class-specific failure modes.

V. Results

Table II shows epoch-level results. The best validation macro F1 of 0.862 was achieved at epoch 5 (* best checkpoint). The full experiment log is publicly available at wandb.ai/didem-demir_01-middle-east-technical-university/DI725-sentiment under run bert-tabular-v1.

Epoch	***Tr.Loss***	***Tr.F1***	**Val.Loss**	***Val.F1***
1	1.018	0.435	0.597	0.716
2	0.501	0.772	0.400	0.827
3	0.264	0.907	0.487	0.841
4	0.142	0.954	0.797	0.805
5	0.089	0.969	0.591	0.862
6	**0.055**	**0.988**	**0.765**	**0.831**

Table II. Training and validation results per epoch

Class	***Precision***	***Recall***	**F1**
Neutral	0.50	0.90	0.64
Negative	0.78	0.70	0.74
Positive	1.00	0.30	0.46
Macro	0.76	0.63	0.614

Table III. Per class test results

The model demonstrates strong performance on the negative (F1 = 0.74) and neutral (F1 = 0.64) classes but exhibits low recall for the positive class (0.30), misclassifying six of ten positive test samples as neutral. This limitation is attributable to the scarcity of positive training examples, even after augmentation. Positive precision is perfect (1.00), indicating a conservative and reliable decision boundary. The gap between the best validation F1 (0.862) and test F1 (0.614) reflects both the small test set (30 samples) and distributional shift resulting from augmented training data. Tail truncation proved effective, as conversation endings contain sentiment-rich resolution signals. Categorical embeddings provided structural context that facilitated rapid convergence, as evidenced by the substantial F1 gain between epochs one and two.

VI. Conclusion

This work presents BertTabular, a fine-tuned BERT model combined with categorical feature embeddings for three-class customer-service sentiment classification. Key contributions include Cramer's V-guided feature selection to eliminate data leakage, multi-strategy augmentation for the minority positive class, tail truncation for long dialogues, and the integration of text and tabular representations. The model achieves a test macro-F1 score of 0.614, with perfect precision for the positive class. Future research could investigate larger models such as RoBERTa or DeBERTa and conversation-aware architectures that leverage turn-level structure.